Pranav Pappu

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Availability: Jan–Dec 2027 for internships in the San Francisco Bay Area

Singapore citizen; eligible for 12-month J-1 visa sponsored by NUS; program hosted by Stanford University.

Education

National University of Singapore 2025 – 2029

B.Comp. (Hons) in Computer Science • **Teaching Assistant, CS1101S Programming Methodology**

AIE Scholarship – Selected as 1 of 20 recipients

Technical Skills

Languages/Runtime: Python • Zig • C++ • Rust • TypeScript/JavaScript • Node.js
Infrastructure/DevOps: Docker/Compose • GitHub Actions CI/CD • Fly.io • Linux
Backend/Data: FastAPI • PostgreSQL/pgvector • SQLAlchemy/Alembic • Pydantic
AI/Testing: PyTorch • Transformers • OpenAI API • MCP (stdio/HTTP) • pytest • Playwright

Experience

Founder, Tweakler 2026 – Present

- Founded Tweakler, a reference-led website generation system that helps agents avoid generic UI: from a one-line brief, it derives an audience's values and jobs-to-be-done, surfaces diverse permissioned design references, and transforms a selected reference into a distinct, editable site with brand-specific copy, assets, and visual direction. Piloted with studios including Brightsea Studio.
- Built a 54-tool MCP workflow and deterministic browser capture-to-code engine for rights-gated cloning, responsive site generation, transactional copy and asset replacement with rollback, and fail-closed desktop/mobile visual, accessibility, build, and type-check validation.

Research Assistant, National University of Singapore Jan 2026 – Present

- Built a dynamic 3DGS benchmark with 11 YAML pipelines across 6 method repos and 3 datasets, plus a 6-metric evaluator for repeatable quality/performance reports.

CCSGP Fellow (AssessMate), NUS Centre for Computing for Social Good 2025 – Present

- Building AssessMate, an AI case-notes copilot for structured documentation; piloted with 2 assessors across multiple sessions to measure documentation-time reduction on PDPA-sensitive child-development data.

C++ and Python Specialist (Freelance), Outlier.ai – Remote Dec 2024 – May 2025

- Evaluated C++/Python model outputs for correctness, edge-case coverage, and reasoning quality, contributing feedback for code-generation model improvement.

Projects

EvoServe – Harness Evaluation & Quality Control 2026 – Present

- Built EvoServe, a harness-quality and evaluation control plane for coding-agent systems: compares policy candidates, ingests captured runs, independently verifies evidence, and produces auditable reports for quality and regression review.
- Implemented a 10-command validate/replay/score/verify/report CLI with 36 JSON schemas and 24 Zig tests; fails closed on malformed, missing, ambiguous, mutated, or untrusted evidence while preserving explicit review before policy activation.

GPT-2 (124M) Recreation from Scratch 2024

- Recreated GPT-2 124M from scratch (tokenizer, transformer blocks, optimizer, mixed precision, and FineWeb Edu evaluation); trained on 10B tokens.

WikiBase – Source-Grounded Knowledge Platform 2026

- Built WikiBase, a student knowledge platform that turns PDFs and notes into cited Markdown wikis, grounded Q&A, and flashcards; architected its FastAPI/PostgreSQL/pgvector backend, idempotent 5-stage pipeline, Docker/Fly.io release infrastructure, and GitHub Actions CI/CD plus a fail-closed 487-test harness covering migrations, security scans, and smoke checks.

Competitions

Replit Agent Hackathon 2026

- 1st place: Built Remember, an AR memory assistant with 4 AI workflows for face recognition, speech transcription, LLM memory extraction, and voice Q&A over 128-float face embeddings to a persistent personal knowledge graph for context-aware recall across repeated encounters.

Google DeepMind × Cactus (YC S25) Global Hackathon 2026

- 2nd place: Built SecureClaw, a privacy router for 12 PII types with 3 routing modes, on-device/cloud inference, and a 30-case benchmark.